

Linux Monitoring Commands

Networking Commands

1. Check server network interface configuration

- ifconfig

This command displays or configures network interfaces, showing IP addresses, netmasks, and other interface details.

If command “ifconfig” not found, install it using command below.

sudo apt install net-tools

```
ubuntu@ip-172-31-18-14:~$ ifconfig
Command 'ifconfig' not found, but can be installed with:
sudo apt install net-tools
ubuntu@ip-172-31-18-14:~$ sudo apt install net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 204 kB of archives.
After this operation, 811 kB of additional disk space will be used.
Get:1 http://us-east-1-ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Fetched 204 kB in 0s (10.6 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 140930 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-0.1ubuntu4.4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.4.0-1029-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.14.0-1010-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
ubuntu@ip-172-31-18-14:~$ ifconfig
enX0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 9001
    inet 172.31.18.14 netmask 255.255.240.0 broadcast 172.31.31.255
    inet6 fe80::8ff:e5ff:fed2:3945 prefixlen 64 scopeid 0x20<link>
    ether 0a:ff:e5:d2:39:45 txqueuelen 1000 (Ethernet)
    RX packets 135560 bytes 198570746 (198.5 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5939 bytes 541697 (541.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 176 bytes 19427 (19.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 176 bytes 19427 (19.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Test connectivity to thecloudadvisory.com

- `ping -c 4 thecloudadvisory.com`

This command sends four ICMP echo requests to the specified domain to check network connectivity.

```
ubuntu@ip-172-31-18-14:~$ ping -c 4 thecloudadvisory.com
PING thecloudadvisory.com (52.85.132.121) 56(84) bytes of data.
64 bytes from server-52-85-132-121.iad50.r.cloudfront.net (52.85.132.121): icmp_seq=1 ttl=247 time=0.810 ms
64 bytes from server-52-85-132-121.iad50.r.cloudfront.net (52.85.132.121): icmp_seq=2 ttl=247 time=0.936 ms
64 bytes from server-52-85-132-121.iad50.r.cloudfront.net (52.85.132.121): icmp_seq=3 ttl=247 time=1.20 ms
64 bytes from server-52-85-132-121.iad50.r.cloudfront.net (52.85.132.121): icmp_seq=4 ttl=247 time=0.964 ms

--- thecloudadvisory.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.810/0.977/1.201/0.141 ms
```

3. Check open ports and listening services

- `sudo netstat -tulnp`

lists all active listening ports and associated services; if netstat is not installed

```
ubuntu@ip-172-31-18-14:~$ sudo netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN      3099/systemd-resolv
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN      3099/systemd-resolv
tcp        0      0 0.0.0.0:80             0.0.0.0:*               LISTEN      598/nginx: master p
tcp6       0      0 :::22                  :::*                   LISTEN      1/systemd
udp        0      0 127.0.0.1:323          0.0.0.0:*               LISTEN      2512/chronyd
udp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN      3099/systemd-resolv
udp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN      3099/systemd-resolv
udp        0      0 172.31.18.14:68        0.0.0.0:*               LISTEN      3098/systemd-networ
udp6       0      0 :::1:323               :::*                   LISTEN      2512/chronyd
```

4. Perform DNS lookup for pravinmishra.in

- `dig pravinmishra.in`

queries DNS to retrieve information about the domain; alternatively, host pravinmishra.in can be used for a simpler output.

```
ubuntu@ip-172-31-18-14:~$ dig pravinmishra.in

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> pravinmishra.in
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 21692
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;, udp: 65494
;; QUESTION SECTION:
;pravinmishra.in.                IN      A

;; ANSWER SECTION:
pravinmishra.in.                60      IN      A       35.154.180.163
pravinmishra.in.                60      IN      A       35.154.89.202
pravinmishra.in.                60      IN      A       43.205.157.51

;; Query time: 42 msec
;; SERVER: 127.0.0.53#53 (127.0.0.53) (UDP)
;; WHEN: Mon Aug 18 09:51:30 UTC 2025
;; MSG SIZE rcvd: 92
```

5. Download a file using wget

- `wget -O /tmp/Untitled-design-40.png https://pravinmishra.in/wp-content/uploads/2023/08/Untitled-design-40.png`
downloads the specified file and saves it to the specified location.

```
ubuntu@ip-172-31-18-14:~$ wget -O /tmp/Untitled-design-40.png https://pravinmishra.in/wp-content/uploads/2023/08/Untitled-design-40.png
--2025-08-18 09:51:44-- https://pravinmishra.in/wp-content/uploads/2023/08/Untitled-design-40.png
Resolving pravinmishra.in (pravinmishra.in)... 43.205.157.51, 35.154.89.202, 35.154.180.163
Connecting to pravinmishra.in (pravinmishra.in)|43.205.157.51|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1702553 (1.6M) [image/png]
Saving to: '/tmp/Untitled-design-40.png'

/tmp/Untitled-design-40.png      100%[=====] 1.62M  949KB/s  in 1.8s
2025-08-18 09:51:47 (949 KB/s) - '/tmp/Untitled-design-40.png' saved [1702553/1702553]
```

Process Monitoring & Control

1. List all running processes

- `ps -e`
displays a list of all currently running processes on the system.

```
ubuntu@ip-172-31-18-14:~$ ps -e
  PID TTY          TIME CMD
    1 ?        00:00:04 systemd
    2 ?        00:00:00 kthreadd
    3 ?        00:00:00 pool_workqueue_release
    4 ?        00:00:00 kworker/R-rcu_g
    5 ?        00:00:00 kworker/R-rcu_p
    6 ?        00:00:00 kworker/R-slub
    7 ?        00:00:00 kworker/R-netns
   10 ?        00:00:00 kworker/0:0H-events_highpri
   11 ?        00:00:00 kworker/u30:0-events_power_efficient
   12 ?        00:00:00 kworker/R-mm_pe
   13 ?        00:00:00 rcu_tasks_rude_kthread
   14 ?        00:00:00 rcu_tasks_trace_kthread
   15 ?        00:00:00 ksoftirqd/0
   16 ?        00:00:00 rcu_sched
   17 ?        00:00:00 migration/0
   18 ?        00:00:00 idle_inject/0
   19 ?        00:00:00 cpuhp/0
   20 ?        00:00:00 kdevtmpfs
   21 ?        00:00:00 kworker/R-inet_
   23 ?        00:00:00 kauditd
   24 ?        00:00:00 khungtaskd
   25 ?        00:00:00 oom_reaper
   26 ?        00:00:00 kworker/u30:2-events_unbound
   27 ?        00:00:00 kworker/R-write
   28 ?        00:00:00 kcompactd0
   29 ?        00:00:00 ksmd
   30 ?        00:00:00 khugepaged
   31 ?        00:00:00 kworker/R-kinte
   32 ?        00:00:00 kworker/R-kbloc
   33 ?        00:00:00 kworker/R-blkcg
   34 ?        00:00:00 irq/9-acpi
   35 ?        00:00:00 xen-balloon
   36 ?        00:00:00 kworker/R-tpm_d
   37 ?        00:00:00 kworker/R-ata_s
   38 ?        00:00:00 kworker/R-md
   39 ?        00:00:00 kworker/R-md_bi
   40 ?        00:00:00 kworker/R-edac-
   41 ?        00:00:00 kworker/R-devfr
   42 ?        00:00:00 watchdogd
   43 ?        00:00:00 kworker/0:1H-kblockd
   44 ?        00:00:00 kswapd0
   45 ?        00:00:00 ecryptfs-kthread
   46 ?        00:00:00 kworker/R-kthro
   47 ?        00:00:00 kworker/R-acpi_
   48 ?        00:00:00 kworker/R-acpi_
```

2. Search for the nginx process

- `ps aux | grep nginx`

filters the list of all processes to show only those related to nginx.

```
ubuntu@ip-172-31-18-14:~$ ps aux | grep nginx
root        598  0.0  0.1 11156 1468 ?        Ss   09:07   0:00 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
www-data    604  0.0  0.4 12880 4284 ?        S    09:07   0:00 nginx: worker process
ubuntu     11005  0.0  0.2  7076 2048 pts/0    S+   09:54   0:00 grep --color=auto nginx
```

3. View process hierarchy

- `pstree`

shows running processes in a tree structure, illustrating parent-child relationships.

```
ubuntu@ip-172-31-18-14:~$ pstree
systemd───ModemManager───3*[{ModemManager}]
├──acpid
├──2*[agetty]
├──amazon-ssm-agent───6*[{amazon-ssm-agent}]
├──chronyd──chronyd
├──cron
├──dbus-daemon
├──multipathd───6*[{multipathd}]
├──networkd-dispatcher
├──nginx──nginx
├──polkitd───3*[{polkitd}]
├──rsyslogd───3*[{rsyslogd}]
├──snapd───8*[{snapd}]
├──sshd──sshd──sshd──bash──pstree
├──systemd──(sd-pam)
├──systemd-journal
├──systemd-logind
├──systemd-networkd
├──systemd-resolved
├──systemd-udev
├──udisksd───5*[{udisksd}]
└──unattended-upgr──(unattended-upgr)
```

4. Terminate a process (choose a harmless process)

- `kill <PID>`

sends a termination signal to the specified process ID;

use `kill -9 <PID>` for a forceful termination if necessary.(use SUDO if still not permitted)

```
ubuntu@ip-172-31-18-14:~$ kill 11005
ubuntu@ip-172-31-18-14:~$ ps aux | grep 11005
-rpsr: kill: (18) - Operation not permitted
ubuntu@ip-172-31-18-14:~$ kill 11005
-rpsr: kill: (18) - Operation not permitted
ubuntu@ip-172-31-18-14:~$ kill 18
```

System Information

1. Display full system information

- `uname -a`
provides detailed information about the system, including kernel version and architecture.

```
bash: kill: (18) Operation not permitted
ubuntu@ip-172-31-18-14:~$ uname -a
Linux ip-172-31-18-14 6.8.0-1029-aws #31-Ubuntu SMP Wed Apr 23 18:42:41 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
```

2. Check system uptime

- `uptime`
shows how long the system has been running since the last boot.

```
ubuntu@ip-172-31-18-14:~$ uptime
09:57:07 up 49 min, 1 user, load average: 0.00, 0.00, 0.00
```

3. Display currently logged-in users

- `who`
lists all users currently logged into the system

```
ubuntu@ip-172-31-18-14:~$ who
ubuntu pts/0 2025-08-18 09:16 (18.206.107.27)
```

4. Check memory usage

- `free -h`
displays the system's memory usage, including total, used, and free memory, in a human-readable format.

```
ubuntu@ip-172-31-18-14:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           957Mi       355Mi       130Mi       908Ki       665Mi       601Mi
Swap:           0B           0B           0B
```

5. Check disk usage

- `df -h`

shows the amount of disk space used and available on all mounted filesystems.

```
ubuntu@ip-172-31-18-14:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  3.6G  3.2G  53% /
tmpfs            479M   0  479M   0% /dev/shm
tmpfs            192M  892K  191M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
/dev/xvda16      881M  149M  671M  19% /boot
/dev/xvda15      105M   6.2M   99M   6% /boot/efi
tmpfs            96M   12K   96M   1% /run/user/1000
```

6. Check directory size usage in /var

- `sudo du -sh /var/*`

estimates and displays the disk usage of each subdirectory within /var.

```
ubuntu@ip-172-31-18-14:~$ sudo du -sh /var/*
1.3M  /var/backups
143M  /var/cache
4.0K  /var/crash
551M  /var/lib
4.0K  /var/local
0     /var/lock
27M   /var/log
4.0K  /var/mail
4.0K  /var/opt
0     /var/run
52K   /var/snap
16K   /var/spool
48K   /var/tmp
1.2M  /var/www
ubuntu@ip-172-31-18-14:~$
```

i-027fdf4ef594d2f8e (ReactApp_Ubuntu)

PublicIPs: 54.234.76.145 PrivateIPs: 172.31.18.14

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences